

SUBSTRATE (Tech-ThreeJs)

Scroll-driven journey across nine orders of magnitude

★ MASTERS STANDOUT	Portfolio	Proud	Masters	Production
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Repo / Path	asadbek/3Js/Tech-ThreeJs
Domain	Creative WebGL / scrollytelling
Stack	Three.js, TypeScript, Vite, GSAP, Lenis, custom circuit-trace GLSL shader, UnrealBloom
Status	Production - ~88% complete
Live / Deploy	Not publicly deployed

Overview

Scroll equals zoom: the scene travels across ~9 orders of magnitude - transistor -> chip die -> device stack -> network -> planet-scale data field - assembling at each rung.

Key Features & Technical Highlights

- Custom circuit-trace GLSL shader; fully procedural (no downloaded models).
- Instanced 1,400-node network rendered in a single draw call.
- 'Scale Gauge' + 'Lab Instrument' design system; 19-file QA harness.

Engineering Evidence & Metrics

- 7 files, ~1,122 LOC; instanced-everything craft with original shaders.

Why It Matters (application / interview talking points)

- Best of the scroll experiences for original shader work and rendering efficiency.